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Duration	6 mins 39 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search stops
b. The list is sorted
c. The search continues from the beginning
d. The search continues in the left or right sublist ✓

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Sorting the list before searching
b. Checking each element sequentially ✓
c. Dividing the list in half repeatedly
d. Skipping every second element

Question **3**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n/2
b. n ✓
c. log n
d. n-1

Question **4**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

- a. O(log n) ✓
- b. O👎
- c. O(1)
- d. O(n log n)

Question **5**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

- a. Binary search does not require dividing the list
- b. Binary search can find multiple instances of the target element
- c. Binary search works on unsorted lists
- d. Binary search has a lower time complexity on large, sorted lists ✓

Question **6**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

- a. Determining whether an element is present in a list ✓
- b. Sorting elements in a list
- c. Inserting elements into a list
- d. Deleting elements from a list

Question **7**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

- a. Selection
- b. Complexity
- c. Insertion
- d. Bubble ✓

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Binary search algorithm is not efficient when the data elements more than 1500 ✓

b. Must use a sorted array

c. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

d. There must be a mechanism to access middle element directly

Question **9**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the list contains duplicate elements

b. When the list is sorted

c. When the element is found in the list ✓

d. When the search algorithm finishes

Question **10**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By comparing each element in the list sequentially ✓

b. By sorting the list first

c. By dividing the list into halves

d. By using a hash function

Question **11**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Hashing

b. Linear search ✓

c. Jump search

d. Binary search

Question **12**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

- a. In the left sublist ✓
- b. In the right sublist
- c. In the entire list
- d. At the beginning of the list

Question **13**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

- a. Linear
- b. Hash
- c. Both (1) & (3)
- d. Binary ✓

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

- a. Linear search requires the list to be sorted.
- b. Linear search can be applied to both sorted and unsorted lists. ✓
- c. Linear search is more efficient than binary search on large lists.
- d. Linear search divides the list into halves.

Question **15**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

- a. Item is the last element in the array or item is not there at all
- b. When the item is the last element in the array
- c. When the item is not the array at all
- d. When the item is somewhere in the middle of the array ✓